FEWLLM Data Agent

KDD Cup 2026 DataAgent-Bench — Course Project

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Society Hub

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Outline

Task & Scoring Rule

Input: `task.json` + heterogeneous context
(CSV, SQLite, JSON, Markdown rules, record-text prose, images)

Output: `prediction.csv` scored by **column-content signature matching**

- Column names **ignored**; row order **ignored**
- Multi-column match via bipartite augmenting paths
- Normalize: numeric $\to 10^{-2}$, string `strip+lower`

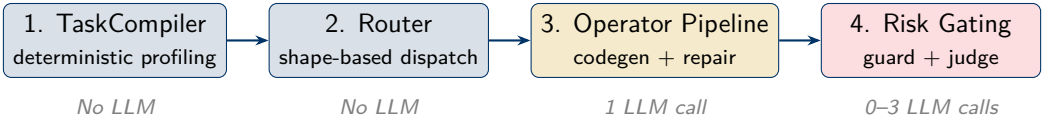
$$\text{score} = \max\left(0,\ \text{recall} - \lambda \cdot \frac{\text{extra\_cols}}{\text{pred\_cols}}\right), \quad \lambda = 0.5$$

Design implications

- **Fewer columns** — every extra wrong col is penalized
- **Preserve granularity** — `first_name+last_name` beats merged `name`
- **Match dtypes** — `163109.0` $\neq$ `163109`

Why Layered? — Four-Layer Architecture

Vanilla ReAct breaks down: easy tasks waste tokens on multi-turn tool calls; hard tasks drift in intermediate steps; no structural guarantee on output shape; no cheap way to verify semantic correctness.



Design principle

Simple tasks pay minimal LLM cost; only suspicious answers escalate to expensive semantic verification.

TaskCompiler & Router — Dispatch by Shape, Not Difficulty

TaskCompiler (zero LLM):

- Per-file scan → `SourceCapability`
  headers, dtypes, cardinality, samples
- Task-level aggregation → `ExecutionProfile`
  - `context_size`, `source_shape`
  - `verifiability`, `operation_complexity`
  - `semantic_rule_required`
- Single source of truth — reused by prompt, static checker, guard, judge

Router rules (hard, deterministic):

- `weak` / `image` → multi_agent
- `long_docs` → RAG-enabled operator
- `semantic_rule_required` → tool_first_mixed
- `small+programmatic+low` → easy

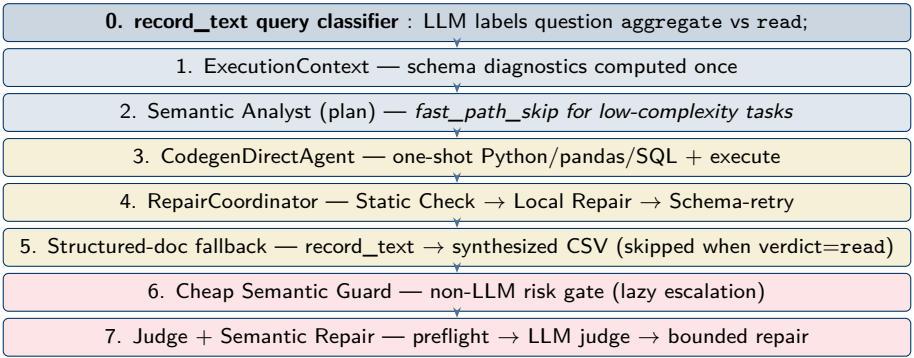
Cascade by *failure type*, not difficulty:

zero_rows → no upgrade (column wrong, not route)
retrieval_empty → RAG route
semantic_consistency_failed → tool_first_mixed
syntax/static/exec error → sibling operator

Cross-model verify (optional)

Primary + verifiers run in parallel; keep columns whose signatures agree $\geq$ `min_agreement` — directly targets the $-\lambda\cdot$ extra penalty.

Operator Pipeline — 8 Stages



Shared evidence — debug_steps

Every generated program emits `schema_inspection`, `used_columns`, `filter_conditions`, `join_keys`, `intermediate_counts`. Read by 3 consumers: cheap guard, preflight, LLM judge — single audit trail, no drift.

Schema Grounding + Static Checker + Local Repair

Schema Grounding

- Extract question concepts
- Match to real columns: Levenshtein + value match + cardinality
- Top-3 candidates w/ samples
- Fights hallucination

Static Checker

- Python AST + SQL scan
- `no_such_column`, `bad_join_key`, `_x`/`_y` suffixes
- Runs *before* execution
- Blocks predictable failures

Local Repair (0 LLM)

- 10+ deterministic patches
- Fuzzy column-name fix
- Cast merge keys to string
- Append missing `answer`
- Zero-row filter normalize

Why these three work together

All three read the **same** `SourceCapability` scan. The prompt tells the LLM what's real; the static checker verifies the LLM didn't invent; local repair fixes the remaining deterministic mistakes without an extra LLM call.

Cheap Semantic Guard — Non-LLM Risk Gate

Position: codegen succeeded, answer structurally valid, no analyst ran yet → trust this answer?

$$\text{risk\_score} = \min\left(1.0, \sum \text{unique\_weights}\right), \quad \text{escalate} = \text{any error} \vee \text{score} \geq 0.5$$

Category	Risk Code	Wt	Meaning
Execution	`execution_failed / empty_answer_rows`	1.0 / 0.75	No/zero answer rows
Repair	`suspicious_fallback:schema_retry`	0.8	Needed schema retry
Trace	`trace_missing:schema_inspection`	0.8	No real-column check
Trace	`zero_intermediate_count`	0.8	Mid-pipeline drops to 0
Coverage	`field_coverage_mismatch`	**0.75**	**Grounding candidate unused**
Formula	`formula_without_trigger`	0.75	Unasked formula used

Killer example

Q: *"patients with severe degree of thrombosis"*. Code uses `disease=="thrombosis"` but grounding found `degree_of_thrombosis` → `field_coverage_mismatch` (0.75) → escalate to judge instead of shipping wrong answer.

Lazy Escalation + Analyst-Exception Recovery

Decision flow:

- Codegen OK, answer valid
- Guard score ≥ 0.5?
  - **No** → fast return (0 extra LLM)
  - **Yes** → `plan(force=True)`
- Plan OK? Yes → Judge + Repair (max 3 rounds)
- Plan failed → fallback plan from guard risks → Judge

Observability — `final_gate` values:

- `plan+judge` — strong path
- `escalated+judge` — recovered
- `cheap_guard_only` — investigate
- `bypassed` — no semantic check

Analyst-exception recovery:

If analyst raises during plan, we used to fall through to a silently-skipped judge. Now:

- `stage==analyst_exception` → retry once with `force=True`
- Still fails → low-confidence synthetic plan; judge must run

Effect

Simple tasks: **1 LLM call total** (codegen only).

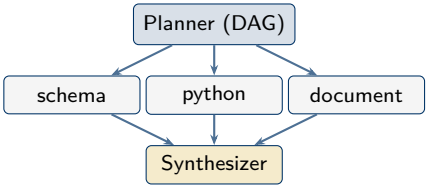
Hard tasks where analyst fails: guaranteed judge engagement instead of silent skip.

RAG Pipeline & Multi-Agent Orchestrator

RAG (long_docs route):

- Heading-aware chunking (1200 chars, overlap 150)
- **Query expansion**: paraphrases + HyDE
- **Hybrid retrieval**: BM25 || Embedding
- **RRF fusion** (rank-only, no score calibration)
- Optional cross-encoder reranker
- Embedding cache: `SHA256(model, text)`

Multi-agent (extreme route):



Specialists only see `report`; only Synthesizer can call `answer`. Same-layer specialists run in parallel.

Self-Consistency, Budget & Demo

Self-consistency — column-signature voting:

- N passes at T=0.7 with different seeds
- `target_count` = mode of column counts
- Top-`target_count` signatures by frequency = winning set
- Pick sample with max coverage, min extras
- Directly targets recall $- \lambda \cdot$ extra/pred

Budget controller (per task):

- LLM / tool / time caps
- Local & reasoner repair caps
- Multi-agent fallback cap

Streamlit demo:

- Real-time subprocess log (ProgressLogger)
- **Reasoning timeline**: schema $\rightarrow$ route $\rightarrow$ codegen $\rightarrow$ guard $\rightarrow$ judge
- Final answer table + generated Python
- Full `trace.json` (collapsible)

```
uv run streamlit run \
  scripts/demo_app.py
```

Auditable, explainable, reproducible.

Summary & Future Work

What we built:

- Tool-first Data Agent pipeline
- Deterministic task profiling (zero LLM)
- Shape-based routing + failure-type cascade
- 10+ local repair rules (zero LLM)
- Cheap Semantic Guard — non-LLM risk gate
- Analyst-exception recovery + audit trail
- Production-grade RAG (BM25+Emb+RRF)
- Multi-agent DAG with bounded replan
- Column-signature self-consistency

Observed performance (public):

Easy: ∼70–80 Medium/Hard: ∼60

Future work:

- Vision executor for image tasks
- Guard-weight auto-calibration from trace FP/FN
- Per-value majority voting within columns
- Smarter record-text extraction for long prose
- Submit to hidden test after tuning

Thank You

Questions?

Code: github.com/zuelmaruixin/kdd